

Repaired Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$:

Items 4–8 of the Round-5 Review

A delta to the 31-page round-4 proof [1],
companion to the (M6') v2 delta [3]

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Disclaimer. This document was drafted by Claude (Anthropic), prompted by Michael Wessel. The results have not been independently verified by human domain experts. We invite scrutiny and corrections.

Abstract

The round-5 review of the 31-page repaired split-forest decidability proof [1] for $\mathcal{ALCI}_{\text{RCC5}}$ [2] lists eight recommended repairs. Item 1 (TeX/PDF mismatch) is editorial and already self-consistent in the repository; items 2, 3 and 9 are addressed by the (M6') recursive-verticalization v2 delta [3]. This delta paper closes the remaining mathematical items 4–8:

4. *Non-circular mosaic patchwork.* We replace the implicit “support against $\mathcal{T}_Q$ ” clause (SC4) of round 4 by an explicit two-step construction: first, a finite abstract pair-label function $L_Q : \text{Pos}_Q^2 \rightarrow \mathbf{B}$ is declared as certificate data; second, an explicit finite validity clause (V11) requires every abstract triple in Trip_Q to satisfy the RCC5 composition table. The mosaic patchwork theorem is then a non-circular consequence.
5. *Strengthened overlap amalgamation.* We strengthen hypothesis (O3) of round 4 to require a single cross-label assignment $L_{12} : (P_1 \setminus P_0) \times (P_2 \setminus P_0) \rightarrow \mathbf{B}$ compatible with all overlap nodes and all mixed triples simultaneously. This becomes an additional explicit validity clause (V12) on certificates.
6. *Joint equality validity clause.* We promote round-4 assumption (A4) of the typed-EQ quotient theorem to an explicit finite validity clause (V13): for every equality class E of ports and every $\exists R.D$ common to the type of E 's members, there is a declared mate/target pair that works uniformly for all ports in E .
7. *Half-open cycle convention.* We rewrite the request-fulfilled repetition lemma with the half-open $[i, j)$ convention. The pumped path is $u_0, \dots, u_{i-1}, (u_i, \dots, u_{j-1})^\omega$, and u_j is identified with the next lap's copy of u_i at the profile level only. The endpoint case is handled explicitly.
8. *Global finite regularization.* We add a global representative-selection lemma: for the finitely many (profile, request-status) classes $\hat{\Pi} \in \hat{\Sigma}$, one representative finite context and one representative request-fulfilled cycle suffices, simultaneously, for every infinite vertical ray of the split forest. The proof is a direct pigeonhole over $\hat{\Sigma}$, not via ω -automata.

None of items 4–8 affects (M6'); the architecture of the v2 delta paper carries over, and the validity-clause numbering is extended from (V1)–(V10) to (V1)–(V13). The decision procedure remains a finite enumeration of finite-graph-checkable certificates; $B(C_0) \leq 2^{2^{p(n)}}$ for a computable polynomial p is preserved.

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1 Scope and notation

This document is a delta to the 31-page round-4 paper [1] and the (M6') v2 delta paper [3]. We use the round-4 notation throughout: Q is a finite generated split-forest certificate; Mos_Q is its support-closed mosaic family; $\mathcal{T}_Q = (\text{Pos}_Q, \text{Pair}_Q, \text{Trip}_Q)$ is the abstract triple graph; $\mathcal{U}(Q)$ is the unfolding; $\equiv_{\text{EQ}}^Q$ is the explicit equality congruence on ports of Q . All other notation is as in [1].

Where the v2 delta paper restated a definition or a clause of [1], we work with the v2 restated form (e.g., recursive verticalization (M6') replaces (M6); validity clauses (V3'), (V4'), (V5'), (V9.Eb') replace (V3), (V4), (V5), (V9.Eb)).

2 Item 4: non-circular mosaic patchwork

2.1 The circularity diagnosed by round 5

Round-4 [1] proves the support-closed mosaic patchwork theorem (Theorem 5.8) in two steps:

1. the mosaic relational layer satisfies the Renz–Nebel hypotheses (RN1)–(RN3) on every mosaic (Lemma 5.5);
2. the overlap-amalgamation lemma (Lemma 5.6) combines two mosaics over a shared boundary;
3. the support-closure clause (SC4) requires that every abstract triple in Trip_Q is represented by some mosaic $M_T \in \text{Mos}$ with matching pair labels (Definition 5.4).

The round-5 review [2] objects that the patchwork theorem uses (SC4) to argue that concrete unfolding triples are RCC5-composition-consistent, while (SC4) itself *presupposes* that the mosaic M_T assigning labels to an abstract triple realises composition-consistent labels. The labelling on $\mathcal{T}_Q$ is implicit, contained in the mosaic data itself rather than declared as an independent function. The risk is that the patchwork claim is established by appeal to mosaic composition consistency, which is in turn what the patchwork claim asserts at the abstract level.

2.2 The repair: explicit L_Q and a composition-table validity clause

We declare an explicit *abstract pair-label function* on the abstract triple graph, separately from any mosaic, and require it to satisfy the composition table. The certificate’s pair labels become primary data; mosaics become local realisations of those labels.

Definition 2.1 (Abstract pair-label function). The certificate $\mathcal{Q}$ declares an explicit function

$$L_Q : \text{Pos}_Q^2 \longrightarrow \mathbf{B}$$

on pairs of abstract position symbols, satisfying:

- (L1) $L_Q(\pi, \pi) = \text{EQ}$;
- (L2) $L_Q(\pi', \pi) = \text{Inv}(L_Q(\pi, \pi'))$;
- (L3) if π, π' are linked by an explicit equality-port chain in E_Q , then $L_Q(\pi, \pi') = \text{EQ}$.

Definition 2.2 (Compatibility with mosaics). For every mosaic $M = (P, \tau, L, \mathcal{V}) \in \text{Mos}_Q$ and every pair $p, q \in P$ realising abstract position symbols $\pi_p, \pi_q \in \text{Pos}_Q$:

$$L_M(p, q) = L_Q(\pi_p, \pi_q).$$

That is, mosaic labels are restrictions of the global L_Q to the mosaic’s position set.

The labels in L_Q are part of the certificate $\mathcal{Q}$ and are part of the finite data the validity check enumerates. We now require explicit composition-table checking on $\mathcal{T}_Q$ as a finite validity clause:

(V11) Composition table on $\mathcal{T}_Q$. For every ordered triple $(\pi_1, \pi_2, \pi_3) \in \text{Pos}_Q^3$,

$$L_Q(\pi_1, \pi_3) \in \text{comp}(L_Q(\pi_1, \pi_2), L_Q(\pi_2, \pi_3)).$$

This is a finite quantifier over $|\text{Pos}_Q|^3 \leq 2|S| \cdot (n+1) \cdot |\text{Ctx}_Q| \cdot k_{\text{strat}} \cdot |\text{Pos}_Q|^2$ abstract triples, each requiring a constant-time table lookup.

Lemma 2.3 ((V11) implies (M3) on every mosaic). *If Q satisfies (V11) and every mosaic $M \in \text{Mos}_Q$ is L_Q -compatible in the sense of Definition 2.2, then M satisfies the round-4 mosaic axiom (M3).*

Proof. For $p, q, r \in P$ at positions $\pi_p, \pi_q, \pi_r \in \text{Pos}_Q$, the mosaic labels are $L_M(p, q) = L_Q(\pi_p, \pi_q)$ (and similarly for the other two pairs). By (V11), $L_Q(\pi_p, \pi_r) \in \text{comp}(L_Q(\pi_p, \pi_q), L_Q(\pi_q, \pi_r))$, so $L_M(p, r) \in \text{comp}(L_M(p, q), L_M(q, r))$, which is (M3). $\square$

2.3 Restated support-closure

The round-4 support-closure clauses (SC1)–(SC4) are restated with explicit reference to L_Q and (V11) replacing the implicit composition-on- $\mathcal{T}_Q$.

Definition 2.4 (Support-closed mosaic family, non-circular). A finite family Mos of mosaics is support-closed relative to $(\mathcal{T}_Q, L_Q)$ if it satisfies (SC1)–(SC3) of [1] unchanged, together with:

- (SC4') For every abstract position symbol $\pi \in \text{Pos}_Q$ on the generated context of some source in Ctx_Q , Mos contains a mosaic M_π that includes π as one of its positions, with $\tau_{M_\pi}(\pi) = \lambda(\pi)$ (the type declared by the certificate) and the stratum of π in M_π matching its declared stratum.

The change is that (SC4') no longer requires per-triple coverage of Trip_Q . Instead, the composition table on $\mathcal{T}_Q$ is enforced *globally* by (V11), and (SC4') requires only that each position has *some* mosaic realising its type and stratum. Triple consistency is then a consequence of (V11) and (SC1)–(SC3), without circular appeal to per-triple mosaic coverage.

2.4 Restated patchwork theorem

Theorem 2.5 (Support-closed mosaic patchwork, non-circular). *Let Q be a valid certificate satisfying (V1)–(V13), (M6') on every mosaic in Mos_Q , (V11) on the abstract pair-label function L_Q , and (SC1)–(SC3), (SC4'). Then every finite prefix $F \subseteq \mathcal{U}(Q)$ satisfies the abstract RCC5 frame axioms (F1)–(F3) on all triples of occurrences.*

Proof. We argue independently of the mosaic family on the global relation:

Step 1. Define the unfolding label $\text{lab}(u, v) \in B$ as $L_Q(\pi_u, \pi_v)$, where π_u, π_v are the abstract position symbols of the unfolding occurrences u, v in $\mathcal{U}(Q)$ (each occurrence carries a unique abstract position symbol by the unfolding rule of §4 of [1], possibly via an equality-port chain).

Step 2. (F1) and (F2) follow from (L1) and (L2) of Definition 2.1. For (F3): given any triple (u, v, w) in F , with abstract position symbols (π_u, π_v, π_w) , apply (V11) directly: $\text{lab}(u, w) = L_Q(\pi_u, \pi_w) \in \text{comp}(L_Q(\pi_u, \pi_v), L_Q(\pi_v, \pi_w)) = \text{comp}(\text{lab}(u, v), \text{lab}(v, w))$. No mosaic is consulted; the composition check is a finite global validity check on L_Q .

Step 3. The mosaic family $\text{Mos}_{\mathcal{Q}}$ is used only to discharge existential support obligations (M4)–(M5) and the EQ-congruence/universal-propagation conditions, which the round-4 lemmas establish on the mosaic side (Lemmas 5.7, 5.8, 5.9 of [1]). These do not rely on (SC4) per-triple coverage; they are local to a single mosaic. Mosaic type-consistency at each abstract position is supplied by (SC4'). $\square$

Remark 2.6. The repair shifts the burden of composition consistency from mosaic-mediated (SC4) coverage to a global validity clause on the finite $L_{\mathcal{Q}}$. The proof of (F3) no longer requires the existence of a mosaic per triple, and no statement of the patchwork theorem assumes its own conclusion.

3 Item 5: strengthened overlap amalgamation

3.1 The weakness diagnosed by round 5

Round-4's overlap-amalgamation lemma (Lemma 5.6 of [1]) takes hypothesis (O3):

every triple (u, v, w) with $u \in P_1 \setminus P_0$, $w \in P_2 \setminus P_0$, $v \in P_0$ admits some $\ell \in \mathbf{B}$ with $\ell \in \text{comp}(L_1(u, v), L_2(v, w))$.

The construction then takes $L_{12}(u, w) \in \bigcap_{v \in P_0} \text{comp}(L_1(u, v), L_2(v, w))$. Round 5 [2] notes that per-triple existence of some ℓ does not imply that the intersection is non-empty, nor that one consistent assignment exists across all cross-pairs simultaneously.

3.2 Strengthened hypothesis (O3') and validity clause (V12)

Definition 3.1 (Strong overlap compatibility). Two mosaics $M_1, M_2 \in \text{Mos}_{\mathcal{Q}}$ are *strongly overlap-compatible* over $P_0 = P_1 \cap P_2$ if there exists a declared cross-label function

$$L_{12}^{\text{cross}} : (P_1 \setminus P_0) \times (P_2 \setminus P_0) \longrightarrow \mathbf{B}$$

satisfying:

(O3'.a) **Triangle compatibility through every overlap node.** For every $u \in P_1 \setminus P_0$, $w \in P_2 \setminus P_0$, and every $v \in P_0$,

$$L_{12}^{\text{cross}}(u, w) \in \text{comp}(L_1(u, v), L_2(v, w)).$$

(O3'.b) **Composition consistency on mixed triples.** For every $u, u' \in P_1 \setminus P_0$, $w \in P_2 \setminus P_0$,

$$L_{12}^{\text{cross}}(u', w) \in \text{comp}(L_1(u', u), L_{12}^{\text{cross}}(u, w)),$$

and symmetrically for $u \in P_1 \setminus P_0$, $w, w' \in P_2 \setminus P_0$.

(O3'.c) **Inverse symmetry.** $L_{12}^{\text{cross}}(w, u) = \text{Inv}(L_{12}^{\text{cross}}(u, w))$.

(V12) Declared overlap labels. For every pair of mosaics $M_1, M_2 \in \text{Mos}_{\mathcal{Q}}$ that share a non-empty boundary $P_0 = P_1 \cap P_2$ and whose amalgamation is required by (SC3), the certificate declares a cross-label function L_{12}^{cross} satisfying (O3'.a)–(O3'.c) of Definition 3.1. In particular, the

declared L_{12}^{cross} extends consistently to a global mosaic $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12})$ with $L_{12} \subseteq L_Q$ (consistent with Definition 2.2).

The clause (V12) is finite-checkable: the certificate enumerates the pairs of mosaics requiring amalgamation (finite, since Mos_Q is finite), and the cross-label function for each pair is a finite map. The composition-table checks of (O3'.a)–(O3'.c) are finite quantifiers over $|P_1 \setminus P_0| \cdot |P_0| \cdot |P_2 \setminus P_0|$ triples, each a constant-time lookup.

3.3 Restated overlap amalgamation lemma

Lemma 3.2 (Overlap amalgamation, strong). *Let $M_1, M_2 \in \text{Mos}_Q$ be strongly overlap-compatible (Definition 3.1), and let L_{12}^{cross} be the declared cross-label function from (V12). Then the amalgamated mosaic $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12}, \mathcal{V}_{12})$ with*

$$L_{12}(p, q) := \begin{cases} L_i(p, q) & \text{if } p, q \in P_i, \\ L_{12}^{\text{cross}}(p, q) & \text{if } p \in P_1 \setminus P_0, q \in P_2 \setminus P_0, \\ \text{Inv } L_{12}^{\text{cross}}(q, p) & \text{if } p \in P_2 \setminus P_0, q \in P_1 \setminus P_0, \end{cases}$$

satisfies (M1)–(M5) and (M6') of v2 [3] on all positions of $P_1 \cup P_2$.

Proof. (M1) and (M2) follow from (O3'.c) and the unchanged structure on P_1, P_2 . (M3) on a triple (u, v, w) in P_{12} :

- If u, v, w all lie in P_i for some i , (M3) on M_i suffices.
- If two of u, v, w lie in P_i and the third in $P_j \setminus P_0$ ($i \neq j$), apply (O3'.b) to the cross pair through the third overlap vertex (or one of the within- P_i vertices), which is a triangle of the form covered by (O3'.a) or (O3'.b).
- If $u \in P_1 \setminus P_0, v \in P_0, w \in P_2 \setminus P_0$, (O3'.a) directly gives $L_{12}^{\text{cross}}(u, w) \in \text{comp}(L_1(u, v), L_2(v, w))$.
- All other cases reduce to (O3'.b) by symmetry.

(M4) and (M5) are inherited from M_1, M_2 as in round-4 Lemmas 5.7 and 5.8. (M6') is preserved: (M6'.b) requires co-vertical PP/PPI labels; cross-pair labels chosen by L_{12}^{cross} are in $\{\text{DR}, \text{PO}\}$ on cross-stratum pairs because (O3') does not produce PP/PPI across distinct strata. (M6'.c)–(M6'.d) are similarly preserved. $\square$

Remark 3.3. Round-4's existential “some $\ell \in \mathbf{B}$ ” in (O3) is replaced by an explicit declared function L_{12}^{cross} . The ambiguity “does the intersection over $v \in P_0$ have a common witness?” is moved to the certificate-writing time: the writer must produce a witness function and (V12) checks it.

4 Item 6: joint equality validity clause

4.1 The diagnosis

Round-4 [1] proves the standalone typed-EQ quotient theorem 7.3 under six assumptions (A1)–(A6). The delicate one is (A4):

For all $u \equiv_{\text{EQ}} v$ and every $\exists R.D \in \lambda(u) = \lambda(v)$ with $R \in \{\text{PP}, \text{PPI}\}$, there exists $w \in \Omega$ with $\{u, v\} \text{Reach}_R w$ (equality-aware reachability) and $D \in \lambda(w)$. Equivalently, a single witness discharges the obligation simultaneously at every $\equiv_{\text{EQ}}$ -mate of the source.

Round-4 derives (A4) from (V4) + the definition of Reach . Round-5 [2] suggests promoting (A4) to an explicit finite validity clause.

4.2 The repair: clause (V13)

(V13) Joint equality discharge. For every mosaic $M \in \text{Mos}_{\mathcal{Q}}$, every stratum $V \in \text{Strat}_M$, every explicit equality class $E \subseteq P_V$ (entirely within V by (M6'.a)), and every $\exists R.D \in \text{cl}(C_0)$ such that $\exists R.D \in \tau(p)$ for all $p \in E$, the certificate declares a joint witness $w_E^{R,D} \in P_V$ with $L_M(p, w_E^{R,D}) = R$ and $D \in \tau(w_E^{R,D})$ for some (equivalently, every) $p \in E$.

In addition, for cross-class joint witnesses, when an equality class $E \subseteq P_V$ has $\exists R.D$ in the common type and the existing port-indexed mate cluster $\text{Mate}_{\mathcal{Q}}(C_V, s_V, p)$ for $p \in E$ contains a port (s', p') targeting a witness in another generated context, the certificate declares that $\text{Mate}_{\mathcal{Q}}(C_V, s_V, p)$ is identical for every $p \in E$ (the cluster is class-wide, not per-port-of-the-class).

4.3 (V13) implies (A4)

Lemma 4.1 ((V13) discharges (A4)). *Suppose $\mathcal{Q}$ satisfies (V1)–(V13). Then the lift of $\equiv_{\text{EQ}}^{\mathcal{Q}}$ to $\mathcal{U}(\mathcal{Q})$ satisfies assumption (A4) of the round-4 typed-EQ quotient theorem.*

Proof. Take $u, v \in \Omega(\mathcal{U}(\mathcal{Q}))$ with $u \equiv_{\text{EQ}} v$ and $\exists R.D \in \lambda(u) = \lambda(v)$ with $R \in \{\text{PP}, \text{PPI}\}$. By the unfolding rule, u and v realise positions p_u, p_v of some mosaic $M \in \text{Mos}_{\mathcal{Q}}$ with p_u, p_v in the same explicit equality class E of M . By (M6'.a), $E \subseteq P_V$ for some stratum V . Since $u \equiv_{\text{EQ}} v$ at the certificate level, both p_u and p_v belong to the same class E .

By (V13), $\mathcal{Q}$ declares a joint witness $w_E^{R,D} \in P_V$ with $L_M(p, w_E^{R,D}) = R$ and $D \in \tau(w_E^{R,D})$ for every $p \in E$. Take w to be the occurrence in $\mathcal{U}(\mathcal{Q})$ realising $w_E^{R,D}$. Then $L_M(p_u, w_E^{R,D}) = R$ and $L_M(p_v, w_E^{R,D}) = R$ both hold, so $u \text{Reach}_R w$ and $v \text{Reach}_R w$ via the same witness occurrence. This is exactly (A4). $\square$

Remark 4.2. The strengthening eliminates the round-4 inference chain through the equality-aware reachability Reach . Where round 4 used the *existence* of a witness for each individual $\equiv_{\text{EQ}}$ -mate, (V13) declares the witness once for the entire class. This is the formal counterpart of the round-5 review’s recommended repair: “promote (A4) to an explicit finite validity clause”.

5 Item 7: half-open cycle convention

5.1 The off-by-one risk in the round-4 convention

Round-4 [1] states the request-fulfilled repetition lemma (Lemma 6.12) with the pumped path

$$u_0, u_1, \dots, u_{i-1}, (u_i, \dots, u_{j-1})^\omega.$$

The cycle word is $u_i, u_{i+1}, \dots, u_{j-1}$; position u_j is not part of the cycle word, but acts as the “return-to- u_i ” on the next lap. Round-5 [2] notes that round-4 alternates between a closed segment $u_i, \dots, u_j$ and a lasso that repeats $u_i, \dots, u_{j-1}$, which is the standard off-by-one source of confusion. The lemma is correct; the convention should be made explicit.

5.2 The repair: half-open $[i, j)$ convention with explicit endpoint mapping

Definition 5.1 (Half-open cycle convention). The cycle word of a request-fulfilled repetition lemma is the *half-open segment* $u_i, u_{i+1}, \dots, u_{j-1}$ of length $j - i$. The pumped path is the concatenation of $u_0, \dots, u_{i-1}$ with infinitely many fresh copies of the cycle word:

$$u_0, u_1, \dots, u_{i-1}, \underbrace{u_i^{(1)}, \dots, u_{j-1}^{(1)}}_{\text{lap 1}}, \underbrace{u_i^{(2)}, \dots, u_{j-1}^{(2)}}_{\text{lap 2}}, \dots$$

For every $\ell \geq 1$, the occurrence $u_j^{(\ell)}$ *does not appear* in the pumped path; the next position after $u_{j-1}^{(\ell)}$ is $u_i^{(\ell+1)}$. At the profile level, $u_j^{(\ell)}$ is identified with $u_i^{(\ell+1)}$ because $\Pi(u_j) = \Pi(u_i)$ by the pigeonhole choice of $i < j$.

Lemma 5.2 (Request-fulfilled repetition lemma, half-open). *Let $u_0, u_1, u_2, \dots$ be an infinite generated vertical path in a split presentation, and suppose every transitive existential request on the path is eventually fulfilled. Then there are indices $i < j$ with $\Pi(u_i) = \Pi(u_j)$ and the half-open segment $[u_i, u_j)$ is request-fulfilled: for every transitive existential request $\exists R.D$ pending at some position u_k with $i \leq k < j$, there is a strict later position $u_{k'}$ within $[u_i, u_j)$ (i.e., $k < k' < j$) or in the segment endpoint u_j (handled below) with $D \in \lambda(u_{k'})$.*

The pumped path $u_0, \dots, u_{i-1}, u_i^{(1)}, \dots, u_{j-1}^{(1)}, u_i^{(2)}, \dots, u_{j-1}^{(2)}, \dots$ built from fresh occurrences of the half-open segment satisfies the strong fulfillment property:

every PP/PPI existential request generated at any occurrence of the pumped path is fulfilled at a strict later occurrence on the same path, within bounded lookahead $\leq j - i$.

Proof. The pigeonhole and checkpoint arguments are unchanged from round-4 (Lemma 6.10 of [1]). We make the endpoint case explicit.

Endpoint case. Let $\exists R.D$ be a request pending at u_k for some $i \leq k < j$. By round-4’s analysis (F1) and (F2a)/(F2b):

- (a) if $\exists R.D$ is fulfilled at some $u_{k'}$ with $k < k' < j$, the pumped path contains $u_{k'}^{(\ell)}$ strictly later than $u_k^{(\ell)}$ in the same lap. Lookahead: $k' - k < j - i$.
- (b) if $\exists R.D$ is fulfilled at u_j in the original path, the pumped path’s next-lap position $u_i^{(\ell+1)}$ has $\Pi(u_i^{(\ell+1)}) = \Pi(u_j)$ at the profile level, but at the *semantic* level $u_i^{(\ell+1)}$ is a fresh occurrence with $\lambda(u_i^{(\ell+1)}) = \lambda(u_i)$, not $\lambda(u_j)$. In general $\lambda(u_i)$ and $\lambda(u_j)$ are equal as types (since profiles include closure types in round-4 Definition 6.6 of [1]), so $D \in \lambda(u_i^{(\ell+1)})$ iff $D \in \lambda(u_j)$. Therefore the pumped path’s $u_i^{(\ell+1)}$ acts as the strict later fulfilling position for $\exists R.D$ generated at $u_k^{(\ell)}$. Lookahead: $(j - k) + 0 = j - k \leq j - i$.

- (c) if $\exists R.D$ persists into $\text{Pend}(u_j) = \text{Pend}(u_i)$, case (F2b) of round-4 applies: there is m with $i \leq m < j$ and $D \in \lambda(u_m)$; the pumped path's $u_m^{(\ell+1)}$ in the next lap satisfies $D \in \lambda(u_m^{(\ell+1)})$, strictly later than $u_k^{(\ell)}$. Lookahead: $(j - k) + (m - i) + 1 \leq 2(j - i)$.

The half-open convention makes case (b) explicit: u_j never appears in the pumped path, but its *profile* is realised by the next lap's $u_i^{(\ell+1)}$, and (since profiles include closure types) the witness label transfers to the next lap. $\square$

Remark 5.3. The change is purely notational: round-4's lemma is correct, but the closed-vs-half-open ambiguity is removed by always using $[i, j)$ and stating the endpoint behaviour explicitly. The boundary u_j is never an occurrence of the pumped path; the position right after the last cycle-word occurrence of a lap is the first cycle-word occurrence of the next lap.

6 Item 8: global finite regularization

6.1 The diagnosis

Round-4 [1] proves the request-fulfilled repetition lemma per-ray (Lemma 6.12 and Lemma 6.13 of [1]): each infinite vertical ray of a split presentation independently lassos to a finite cycle. Round-5 [2] requests a global lemma showing that a single finite certificate covers *all* infinite vertical rays of the branching split forest uniformly, not infinitely many independent lasso choices.

6.2 The repair: a global representative-selection lemma

The key observation is that the finite extended-profile alphabet $\widehat{\Sigma}$ is finite of size at most $|\Sigma| \cdot 2^n$, and a lasso choice depends only on the extended profile $\widehat{\Pi}$ at the two pigeonhole positions, not on the specific ray.

Lemma 6.1 (Uniform representative selection). *Let $\widehat{\Sigma}$ be the finite extended-profile alphabet (round-4 Definition 6.7). For every $\sigma \in \widehat{\Sigma}$, choose:*

- (1) *a representative finite generated context $C_\sigma \in \text{Ctx}_Q$ whose source has extended profile σ ;*
- (2) *a representative cycle word $\text{cyc}_\sigma = (u_0^\sigma, u_1^\sigma, \dots, u_{L_\sigma-1}^\sigma)$ of length L_σ , built from C_σ using the construction of Lemma 5.2, such that the half-open segment cyc_σ is request-fulfilled.*

If, for every $\sigma \in \widehat{\Sigma}$, such a representative $(C_\sigma, \text{cyc}_\sigma)$ exists, then a single finite certificate Q covers every infinite vertical ray of the split forest, by uniformly substituting at every ray the canonical $\text{cyc}_{\sigma(r)}$ for the extended profile $\sigma(r)$ that the ray hits twice.

Proof. We argue in two parts.

Part 1 (Existence of representatives). By the round-4 hypothesis that every transitive existential request is eventually fulfilled in the witness-generated submodel, every infinite vertical ray r satisfies the hypothesis of Lemma 6.10 (checkpoint structure) of [1]. Hence for every r the pigeonhole gives indices $i_r < j_r$ with $\widehat{\Pi}(u_{i_r}^r) = \widehat{\Pi}(u_{j_r}^r) = \sigma(r)$.

For each $\sigma \in \widehat{\Sigma}$ realised on some ray, fix one such ray r_σ and set C_σ to be the generated context of $u_{i_{r_\sigma}}^{r_\sigma}$ and $\text{cyc}_\sigma = [u_{i_{r_\sigma}}^{r_\sigma}, \dots, u_{j_{r_\sigma}-1}^{r_\sigma}]$. This is the *Axiom-of-Choice-free* representative selection: for each of the finitely many extended profiles in $\widehat{\Sigma}$, choose any one ray realising it. Since $\widehat{\Sigma}$ is finite, the selection is finite and effective.

Part 2 (Uniform substitution). Consider any infinite vertical ray r of the split forest. By Part 1, r hits some extended profile $\sigma(r) \in \widehat{\Sigma}$ at two positions $i_r < j_r$ with $\widehat{\Pi}(u_{i_r}^r) = \widehat{\Pi}(u_{j_r}^r)$. By the substitution lemma of [1] (Lemma 6.9), substituting the representative cycle $\text{cyc}_{\sigma(r)}$ for the tail $[u_{i_r}^r, \dots]$ of r preserves all closure formulas at every position of r , because both $(u_{i_r}^r, \dots)$ and $\text{cyc}_{\sigma(r)}$ realise the same extended profile at their starting positions and are request-fulfilled.

The substitution is the *same* cycle word for every ray that hits $\sigma(r)$; rays that hit different extended profiles substitute different (but pre-declared) cycle words. The total substitution data is a finite map $\widehat{\Sigma} \rightarrow \{(C_\sigma, \text{cyc}_\sigma) : \sigma \in \widehat{\Sigma}\}$, finite of size $\leq |\widehat{\Sigma}|$.

After substitution, every infinite vertical ray of the (modified) split forest is the original prefix $u_0^r, \dots, u_{i_r-1}^r$ concatenated with $\text{cyc}_{\sigma(r)}^\omega$. The finitely many cycle words $\{\text{cyc}_\sigma\}_{\sigma \in \widehat{\Sigma}}$ together with the finite prefix data of the unmodified portion form a finite certificate. $\square$

(V14) Uniform cycle representatives. For each $\sigma \in \widehat{\Sigma}$ that is realised on some infinite vertical ray of the witness-generated submodel, the certificate $\mathcal{Q}$ declares one representative finite generated context C_σ and one representative half-open cycle word cyc_σ satisfying the request-fulfilled property of Lemma 5.2. Uniformity: no two distinct extended profiles use the same cycle word, and the substitution function is the identity on the finite prefix $u_0, \dots, u_{i-1}$ before the first pigeonhole hit.

(V14) is a finite quantifier over $|\widehat{\Sigma}| \leq |\Sigma| \cdot 2^n$ extended profiles; for each, the certificate declares a context (one of $|\text{Ctx}_{\mathcal{Q}}|$ many) and a cycle word (a path in $\mathcal{U}(\mathcal{Q})$ of length at most $|\text{Ctx}_{\mathcal{Q}}| \cdot |S|$).

Corollary 6.2 (Global finite regularization). *A valid certificate $\mathcal{Q}$ satisfying (V1)–(V14) yields, by the uniform substitution of Lemma 6.1, a single finite representation of every infinite vertical ray of the original witness-generated submodel. No automaton, parity acceptance, or ω -language theoretic machinery is required; the pigeonhole is over the finite set $\widehat{\Sigma}$.*

7 Summary and combined statement of the proof

The round-5 review [2] lists eight recommended repairs. Combining this paper with the v2 (M6') delta [3] closes all eight:

- **Item 1 (TeX/PDF mismatch).** Self-consistent in the repository (the source compiles to the 31-page PDF; the reviewer's bundle had stale source). Editorial; no mathematical change.
- **Items 2, 3, 9.** Closed by [3]: recursive verticalization (M6'), the stress concept $\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$ accepted by the worked example, polynomial side-width $m(n) = O(n^3)$ replaced by an effectively computable recurrence over modal depth.
- **Item 4.** Closed by §2: explicit abstract pair-label function $L_{\mathcal{Q}}$ and validity clause (V11) for the composition table on $\mathcal{T}_{\mathcal{Q}}$. The patchwork theorem is non-circular.
- **Item 5.** Closed by §3: strong overlap compatibility (O3') and validity clause (V12) requiring a declared global cross-label assignment.

- **Item 6.** Closed by §4: (V13) promotes the joint equality discharge from assumption (A4) to an explicit finite validity clause.
- **Item 7.** Closed by §5: half-open $[i, j)$ convention with explicit endpoint mapping $u_j \mapsto u_i^{(\ell+1)}$ at the profile level.
- **Item 8.** Closed by §6: uniform representative selection (V14) gives a single finite certificate covering all infinite vertical rays.

7.1 Combined validity-clause numbering

The combined certificate validity check has fourteen clauses: (V1)–(V10) of round-4 [1], modified in v2 to (V3'), (V4'), (V5'), (V9.Eb') [3], together with (V11)–(V14) of this paper. Each clause is a finite quantifier over the certificate's finite alphabets and contexts; no clause requires a quantifier over models or unfoldings.

7.2 Combined decision procedure

The decision procedure for $\mathcal{ALCI}_{\text{RCC5}}$ concept satisfiability is:

- (1) enumerate finite generated split-forest certificates $\mathcal{Q}$ of size $\leq B(C_0) \leq 2^{2^{p(n)}}$ for an explicit computable polynomial p , where the components of $\mathcal{Q}$ are: S (profiles), $E_{\mathcal{Q}}$ (equality ports), $\text{Ctx}_{\mathcal{Q}}$ (context schemes), $\text{Mos}_{\mathcal{Q}}$ (mosaics with stratum forests $\mathcal{V}$), $L_{\mathcal{Q}}$ (abstract pair-label function), $\{L_{12}^{\text{cross}}\}$ (declared overlap labels), $\{w_E^{R,D}\}$ (joint equality witnesses), $\{(C_{\sigma}, \text{cyc}_{\sigma})\}_{\sigma \in \widehat{\Sigma}}$ (uniform cycle representatives);
- (2) for each $\mathcal{Q}$, check (V1)–(V14) and (M6'.a)–(M6'.d), all finite-graph computations on $\mathcal{Q}$'s alphabets;
- (3) if any $\mathcal{Q}$ passes, output “ C_0 is satisfiable”; otherwise output “ C_0 is unsatisfiable”.

Soundness is the v2 vertical-realisation lemma [3] combined with Theorem 2.5 and the typed-EQ quotient theorem of round 4 with (A4) replaced by (V13). Completeness is the uniform regularization Corollary 6.2 combined with the existence of representatives from witness-generated submodels of any satisfying abstract RCC5 frame, as in round 4.

References

- [1] GPT-5.5 (OpenAI), prompted by Michael Wessel. *A Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$ Without Automata: A Repaired Containment-Oriented Presentation*. Unpublished manuscript, May 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round2/repaired_split_forest_no_automata_proof.pdf.
- [2] GPT-5.5 (OpenAI), prompted by Michael Wessel. *Formal Review of the Latest Repaired Split-Forest Proof Manuscript*. Unpublished manuscript, May 22, 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round2/formal_review_progress_paper.pdf.

- [3] Claude (Anthropic, Opus 4.7), prompted by Michael Wessel. *Repaired Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$ (v2): Recursive Verticalization ($M6'$) to Close the Round-5 Side-Witness Gap*. Unpublished manuscript, May 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round2/repaired_split_forest_no_automata_proof_v2.pdf.
- [4] Jochen Renz and Bernhard Nebel. *On the Complexity of Qualitative Spatial Reasoning: A Maximal Tractable Fragment of the Region Connection Calculus*. Artificial Intelligence, 108(1–2):69–123, 1999.
- [5] Michael Wessel et al. *ALCI_RCC5 Project Repository*. <https://github.com/lambdamikel/alcircc5>.